AMCIS Q-SEC CORE

Architecture Diagrams

Complete Technical Documentation Package

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Production-Ready Quantum-Secure Security Framework

INTERNAL / COMMERCIAL

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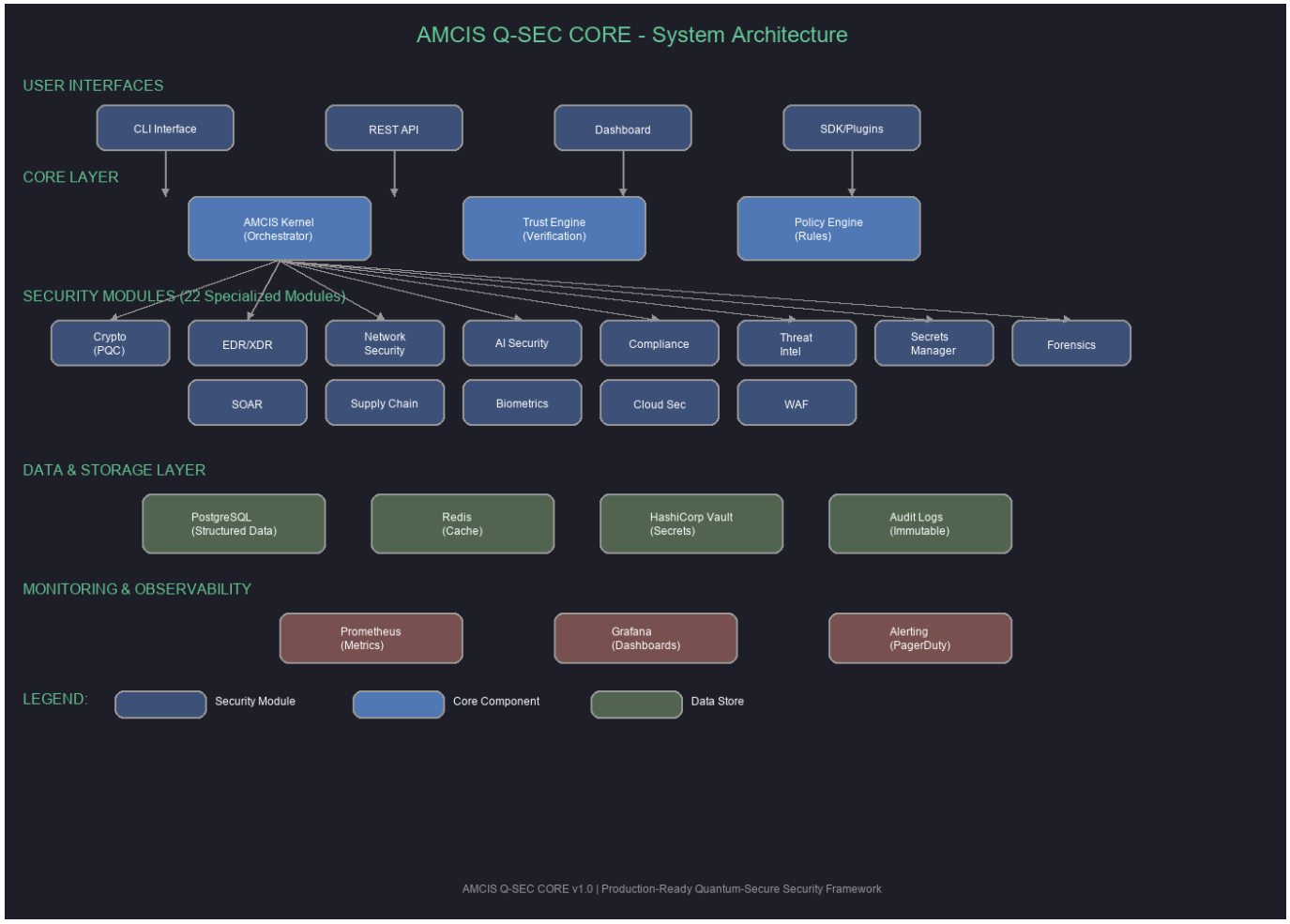
1. System Architecture Overview

The AMCIS system follows a layered architecture with clear separation of concerns:

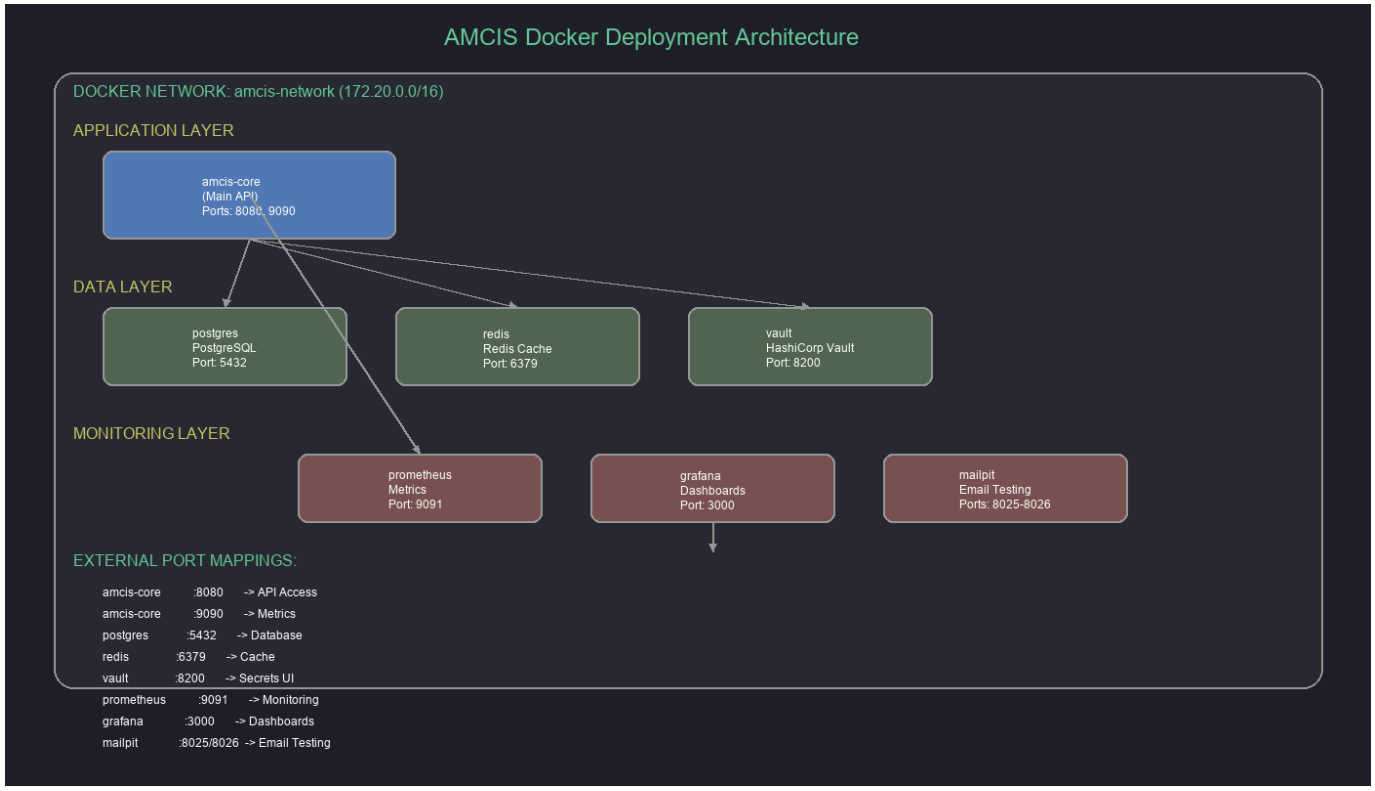
• User Interfaces: CLI, REST API, Dashboard, SDK for different user types
• Core Layer: Central orchestration, trust scoring, and policy enforcement
• Security Modules: 22 specialized modules covering all security domains
• Data Layer: PostgreSQL for structured data, Redis for caching, Vault for secrets
• Monitoring: Prometheus and Grafana for observability



2. Docker Deployment Architecture

AMCIS deploys as a Docker Compose stack with 7 containerized services:

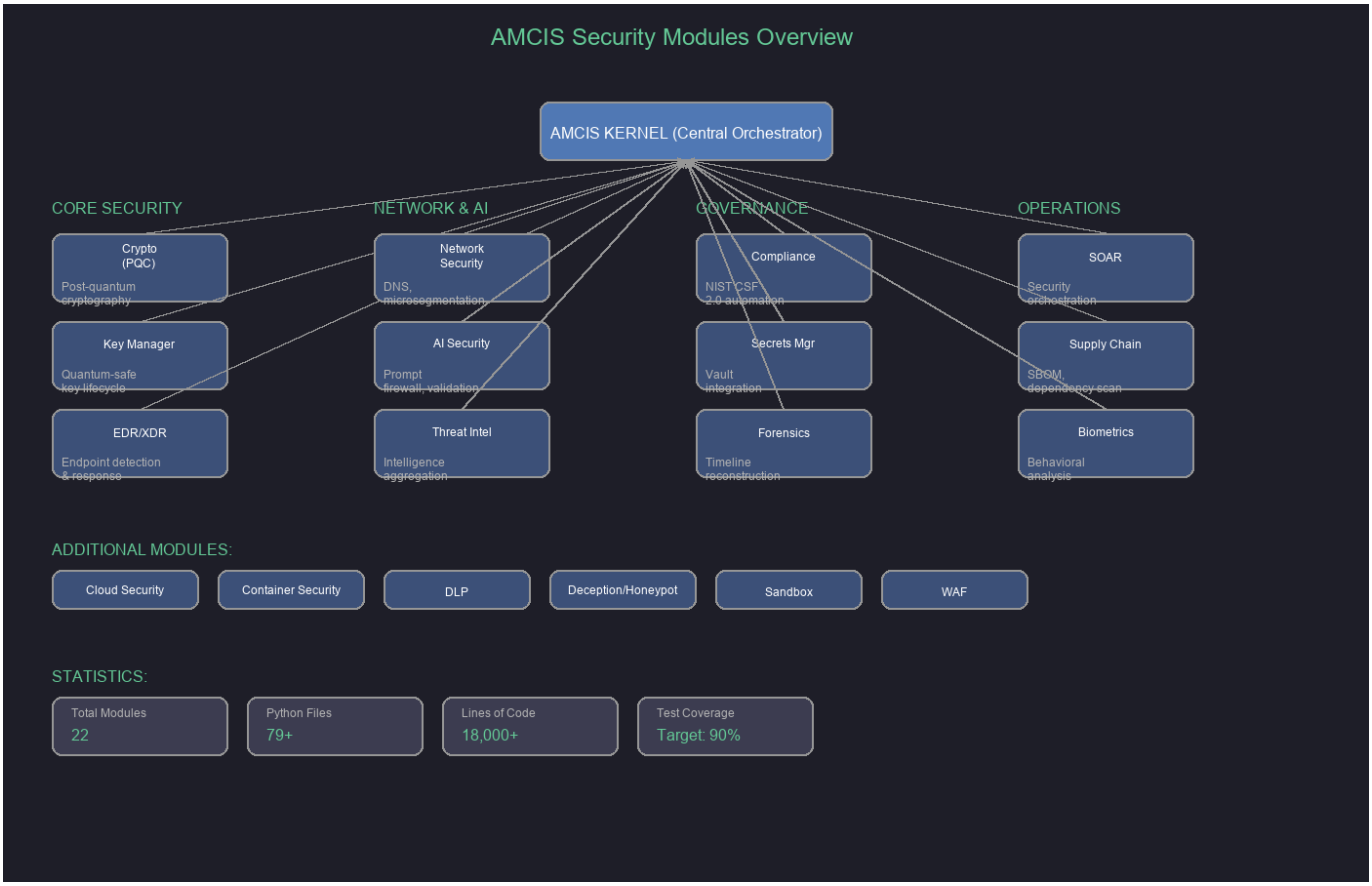
• amcis-core: Main Python application (API on 8080, metrics on 9090)
• postgres: PostgreSQL 15 database for persistent storage
• redis: Redis 7 cache for session management and rate limiting
• vault: HashiCorp Vault for secrets management
• prometheus: Metrics collection and alerting
• grafana: Visualization dashboards
• mailpit: Email testing for development



3. Security Modules Overview

AMCIS includes 22 specialized security modules organized by function:

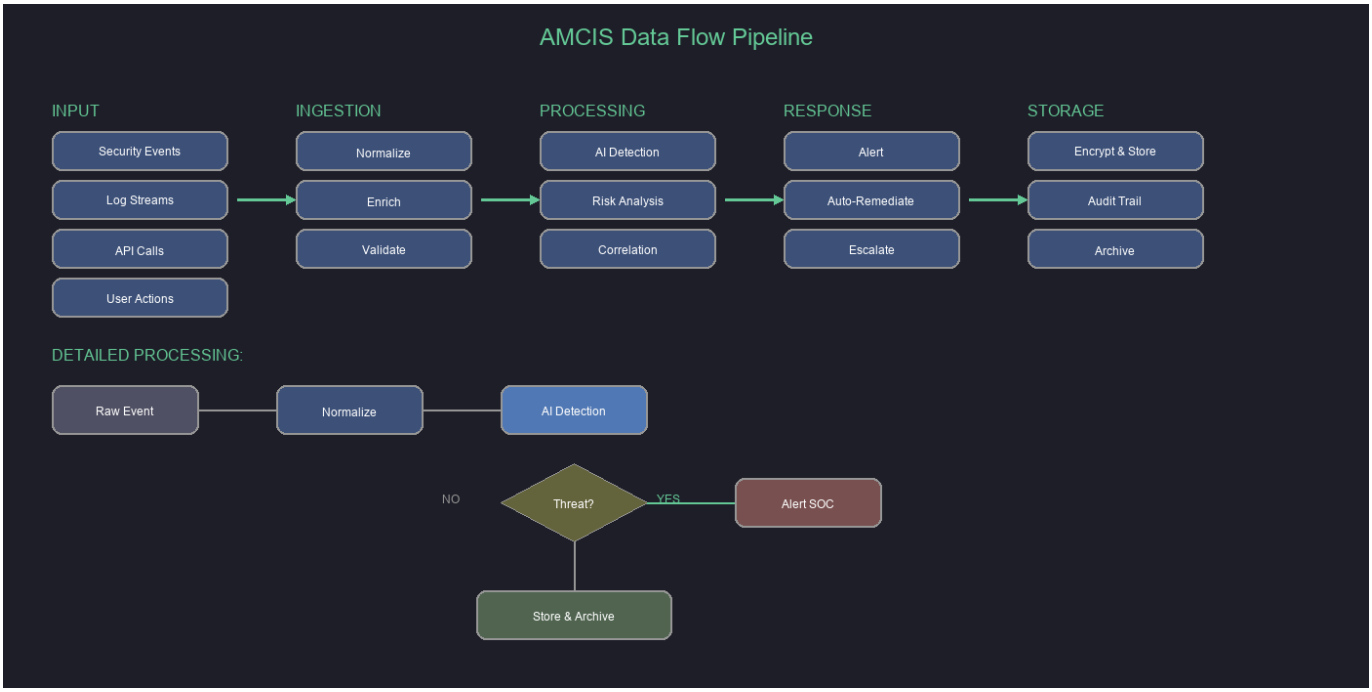
Core Security: Crypto (PQC), Key Manager, EDR/XDR
Network & AI: Network Security, AI Security, Threat Intelligence
Governance: Compliance (NIST CSF), Secrets Manager, Forensics
Operations: SOAR, Supply Chain, Biometrics
Additional: Cloud Sec, Container Sec, DLP, Deception, Sandbox, WAF



4. Data Flow Pipeline

Security data flows through a 5-stage pipeline:

1. INPUT: Security events, logs, API calls, user actions
2. INGESTION: Normalization, enrichment, validation
3. PROCESSING: AI detection, risk analysis, correlation
4. RESPONSE: Alerting, auto-remediation, escalation
5. STORAGE: Encrypted storage, audit trails, archiving



5. Technical Specifications

Specification	Value
Programming Language	Python 3.11+
Framework	FastAPI, Click (CLI)
Total Python Files	79+
Lines of Code	18,000+
Security Modules	22
Database	PostgreSQL 15
Cache	Redis 7
Secrets Management	HashiCorp Vault
Container Platform	Docker + Docker Compose
Monitoring	Prometheus + Grafana
Cryptography	NIST FIPS 203/204 (PQC)
Compliance	NIST CSF 2.0, CMMC, FedRAMP
AI Agents	35+ autonomous agents
Test Coverage Target	90%

Port Mappings:

```
amcis-core API        :8080
amcis-core Metrics    :9090
PostgreSQL            :5432
Redis                 :6379
Vault UI              :8200
Prometheus            :9091
Grafana               :3000
Mailpit SMTP          :8025
Mailpit Web           :8026
```